

Irrationality criteria for Lambert subseries

The series $\sum_{a \in A} (b^a - 1)^{-1}$ and Erdős Problem #257

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ABSTRACT

We prove that $\sum_{a \in A} (b^a - 1)^{-1}$ is irrational for every integer $b \geq 2$ and every infinite $A \subseteq E \cup V$ under two hypotheses. The set E satisfies a finite-prime weighted summability condition at base two, while V admits a summable positive divisor cover. The two support classes are incomparable, both extend beyond reciprocal-summable supports, and the union criterion reaches supports in neither class. The proof turns on a second average over dyadic lengths: it restores the reciprocal factor lost by incomplete residue periods and, because both estimates use one finite distribution, yields a single index at which both displacements are small. The arbitrary infinite-support problem in base two remains open.

1 Introduction

For a set A of positive integers and a real number $b > 1$, write

$$X_A(b) = \sum_{a \in A} \frac{1}{b^a - 1}.$$

This series converges by comparison with a geometric series, since $(b^a - 1)^{-1} \leq b^{1-a} / (b - 1)$. Erdős proved irrationality for full support at integer bases [5], and later proved the same conclusion for pairwise coprime supports of finite reciprocal sum [7, p. 222]. On that page he stated that the coprimality assumption could be removed, without giving the details. The question for arbitrary infinite support in base two is Problem #257:

Problem 1.1 (Erdős #257). Is $X_A(2)$ irrational for every infinite $A \subseteq \mathbb{N}_{>0}$?

The conditions below are hereditary: if a set satisfies one of them, so does each of its infinite subsets. For the first condition, consider an exponent $a = 2^k m$ with m odd. We replace its reciprocal weight $1/(2^k m)$ by $1/[m(b^{2^k} - 1)]$, so that a large power of 2 in a reduces its contribution to the sum. More generally, for a finite set P of primes, let the P -part of a be $h(a) = \prod_{p \in P} p^{v_p(a)}$, where $v_p(a)$ is the exponent of p in a .

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Theorem 1.2 (a weighted condition on the support). *Let $b \geq 2$ be an integer, let $A \subseteq \mathbb{N}_{>0}$ be infinite, and let P be a finite nonempty set of primes. Set $h(a) = \prod_{p \in P} p^{v_p(a)}$. If*

$$W_{b,P}(A) := \sum_{a \in A} \frac{h(a)}{a(b^{h(a)} - 1)} < \infty, \quad (1)$$

then $X_A(b)$ is irrational. In particular, the base-two condition

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty \quad (W)$$

implies that $X_A(b)$ is irrational for every integer $b \geq 2$. Both conclusions are hereditary under passage to infinite subsets.

Consider the support

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, m \leq 2^{2^k}\}.$$

For a fixed k , the sum of $1/m$ over the odd cofactors is of order 2^k . Thus the corresponding terms contribute at least $1/4$ to $\sum_{a \in A_*} 1/a$, but at most 2^{1-k} to (W). The calculation in Section 2 proves both bounds. Every infinite subset of A_* consequently gives an irrational value at every integer base. Since $h/(2^h - 1) \leq 1$, the same theorem also includes reciprocal-summable supports. Section A gives a direct proof of Erdős's stated extension.

The divisor-cover condition instead bounds fractional powers of finite divisor counts by positive sums over divisors (Theorem 3.1). We prove that the two conditions are incomparable and that their union gives a further irrationality criterion: if E satisfies the weighted condition and V has a summable divisor cover, every infinite subset of $E \cup V$ has irrational sum (Theorem 4.1). This conclusion is not obtained by adding two separately irrational subseries: their sum could be rational. What combines are the two small-displacement estimates, once both have been placed on the same finite distribution. The separating constructions in Proposition 5.1 then give a union outside both individual classes (Corollary 5.2). These are sufficient hypotheses for Problem 1.1; the arbitrary support in that problem need not satisfy them.

The proofs compare a multiple of $X_A(b)$ with an integer. For a positive integer N , put

$$\Delta_{b,A}(N) = \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \quad J_{b,A}(N) = \sum_{\substack{a \in A \\ a \leq N}} \sum_{j=1}^{\lfloor N/a \rfloor} b^{N-ja}. \quad (D)$$

The identity follows by division of N by a . If A is infinite, then $\Delta_{b,A}(N) > 0$, since some $a \in A$ exceeds N . Suppose that $b \geq 2$ is an integer and $X_A(b) = p/q$, with $p \in \mathbb{Z}$ and $q \geq 1$ an integer. Then $q\Delta_{b,A}(N)$ is a positive integer, and hence $\Delta_{b,A}(N) \geq 1/q$. To obtain a contradiction, it therefore suffices to make $\Delta_{b,A}(N)$ arbitrarily small. Erdős also proposed using fractional parts of powers times the series value [7, p. 226].

We average over multiples of a modulus divisible by a finite subset of A , whose terms in (D) then vanish. For each remaining exponent, the mean contribution over a

complete residue period is small when its gcd with the modulus is large. The incomplete period causes the difficulty: its bound lacks the factor $1/a$ needed to sum over the exponents. A second average, over dyadic lengths, restores this factor. Indeed, an exponent contributes only at lengths that reach it, and the reciprocals of those lengths have a geometrically decreasing sum. Keeping the dependence on the modulus explicit allows both criteria to use the same finite average.

Section 2 proves the weighted estimate. Section 3 introduces divisor covers through a finite example and proves their averaging estimate. Section 4 combines the criteria. We construct the separating supports in Section 5 and discuss the limits of the argument in Section 6. The main argument ends there. Appendix A gives the inherited reciprocal-summable proof. The other appendices collect finite-denominator results and conditional tests for rational membership, a separate question from the support criteria. Neither $1/2$ nor $1/21$ is decided by those tests.

2 Finite averages and the weighted criterion

Fix the integer base b and the finite prime set P . We average $\Delta_{b,A}$ over multiples of a modulus Q . The mean contribution of an exponent a depends on (Q, a) . For example, when $b = 2$, $Q = 4$ and $a = 6$, the residues are $4, 2, 0$, and

$$\frac{1}{3} \left(\frac{2^4 - 1}{63} + \frac{2^2 - 1}{63} + 0 \right) = \frac{2}{21} \leq \frac{2}{6(2^2 - 1)} = \frac{1}{9}.$$

The final expression is the mean of $2^{4m \bmod 6}/63$ over the same period. If $6 \mid Q$, the term with exponent 6 vanishes at every multiple of Q . In the proof, divisibility removes finitely many exponents, and the gcd estimate bounds the remaining terms.

Proof of Theorem 1.2. The contribution of one exponent. Put $d_a(m) = (b^{m \bmod a} - 1)/(b^a - 1)$ and $g = (Q, a)$. For positive integers Q, a, T , the orbit has length a/g . Splitting $1, \dots, T$ into full orbits and a remainder gives

$$\frac{1}{T} \sum_{t=1}^T d_a(tQ) \leq \frac{g}{a(b^g - 1)} + \frac{1}{T(b^g - 1)}. \quad (2)$$

The sum over one orbit is $\sum_{t=1}^{a/g} b^{tQ \bmod a} / (b^a - 1) = 1/(b^g - 1)$. The first term in (2) accounts for complete orbits; the second allows one extra orbit for the remaining indices. The complete-orbit bound explains the weight in the theorem: replacing g by the P -part $h(a)$ gives its summand. The modulus chosen below will permit this replacement for $h(a) \leq H$; the remaining exponents need a separate estimate. Also, for $Y = QT$,

$$\frac{1}{T} \sum_{t=1}^T \sum_{\substack{a \in A \\ a > Y}} d_a(tQ) \leq \frac{4}{T}, \quad (3)$$

because $d_a(tQ) \leq 2^{tQ-a}$ when $a > QT$ and $\sum_{t=1}^T 2^{tQ-QT} \leq 2$.

Recovering the reciprocal factor. The error in (2) tends to zero for each fixed a , but it has no factor $1/a$ to justify summing over the increasing range $a \leq QT$. We therefore

take $T = 2^j$ and sum first over j . For a fixed a , only lengths $2^j \geq a/Q$ contribute, and the sum of their reciprocals is at most $2Q/a$. Thus, for integers $Q, M \geq 1$ and $\alpha_a \geq 0$ with $\sum_a \alpha_a/a < \infty$,

$$\sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{a \leq Q2^j} \alpha_a \leq 2Q \sum_{a \geq 1} \frac{\alpha_a}{a}; \quad (4)$$

the estimate is unchanged if the available dyadic lengths start later. Dividing by M will turn the cost $2Q$ into $2Q/M$.

Choosing the modulus. Fix $\varepsilon > 0$. Choose finite nonempty $F \subseteq A$ so that the sum in (1) over $A \setminus F$ is below ε , and let L be any fixed positive common multiple of F . Besides cancelling F , the modulus should contain every P -prime power up to a growing threshold H . Truncating each prime power separately keeps its size polynomial in H . With $p_* = \max P$ and $H \geq 2p_*$, set

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \left\lfloor \frac{H}{p_*} \right\rfloor.$$

For $a \in F$ the displacement term vanishes, since $a \mid Q$. For $a \notin F$ with $h(a) \leq H$, we have $h(a) \mid Q$ and hence $(Q, a) \geq h(a)$. The complete-period term in (2) is therefore bounded by $h(a)/[a(b^{h(a)} - 1)]$, using the monotonicity of $n/(b^n - 1)$. The complete-period contributions therefore sum to less than ε . The dyadic estimate will control their incomplete periods.

For $h(a) > H$, we instead have $(Q, a) \geq G$. Sum the complete-period bounds over $a \leq QT$, using $\sum_{a \leq QT} 1/a \leq 1 + \log(QT)$. There are at most QT incomplete-period terms, each at most $1/[T(b^G - 1)]$. The total contribution from these exponents is thus at most

$$\frac{G(1 + \log(QT)) + Q}{b^G - 1}.$$

For the gcd claim, either every P -prime-power component of $h(a)$ is at most H , in which case $h(a) \mid Q$, or some $p^{v_p(a)} > H$ contributes $p^{\lfloor \log_p H \rfloor} > H/p \geq H/p_* \geq G$ to the gcd. *Choosing the number of scales.* A longer dyadic average reduces the incomplete-period cost Q/M . It also increases the largest observation length: since $T = 2^j$ and $j < 2M$, the logarithm in the large-gcd bound contributes a term of order GM/b^G . We must therefore make both Q/M and GM/b^G tend to zero. With F and L fixed, $Q \leq LH^{|P|}$ and $G = H/p_* + O(1)$, so the choice $M = \lfloor b^{G/2} \rfloor$ lies between the two required scales. Here M is the number of dyadic scales, not an observation length. At scale j , the inner average contains $T = 2^j$ sampled multiples, with $M \leq j < 2M$. Combining the preceding estimates with (3), averaging over $T = 2^j$ for $M \leq j < 2M$, and using (4) with $\alpha_a = \mathbf{1}_A(a)/(b^{h(a)} - 1)$ yields

$$\frac{1}{M} \sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{t=1}^{2^j} \Delta_{b,A}(tQ) \leq \varepsilon + \frac{2QW_{b,P}(A)}{M} + \frac{G(1 + \log Q + 2M \log 2) + Q}{b^G - 1} + 4 \cdot 2^{-M}. \quad (5)$$

The four terms respectively bound the weighted tail over complete periods, the incomplete periods, the exponents with large gcd, and the exponents beyond QT . With F and

L fixed, every term after ε tends to zero as $H \rightarrow \infty$. The left-hand side is an average with nonnegative weights summing to one: each of the M lengths has weight $1/M$, distributed uniformly among its 2^j indices. For large H , some $N = Qt$ therefore satisfies $\Delta_{b,A}(N) < 2\varepsilon$. This argument gives no rate of decay in N . Since ε is arbitrary, this contradicts the lower bound $1/q$ under rationality. Finally $W_{b,P}(A) \leq W_{2,P}(A)$ for $b \geq 2$, and the weighted sum decreases on taking subsets. $\square$

2.1 An example beyond reciprocal summability

For the support in the introduction, the two sums can be compared layer by layer. Recall that

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, m \leq 2^{2^k}\}.$$

The layers are disjoint, since their elements have different 2-adic valuations. For $r \geq 2$, put

$$S_r = \sum_{\substack{m \leq 2^r \\ m \text{ odd}}} \frac{1}{m}.$$

Each block $2^j \leq m < 2^{j+1}$, $1 \leq j < r$, contains 2^{j-1} odd integers, with reciprocal sum between $1/4$ and $1/2$. Including $m = 1$ gives $r/4 \leq S_r \leq r$. The sum of reciprocals in the k th layer is $2^{-k} S_{2^k} \geq 1/4$, so $\sum_{a \in A_*} 1/a$ diverges. For $P = \{2\}$, its contribution to the weighted sum satisfies

$$\frac{S_{2^k}}{2^{2^k} - 1} \leq \frac{2^k}{2^{2^k} - 1} \leq 2^{1-k}.$$

The last bound uses $2^{2^k} - 1 \geq 2^{2^k-1}$ and $2^k \geq 2k$. Summation proves (W). The lower bound $1/4$ in every layer has already excluded reciprocal summability.

A related selection of a term from an average over an arithmetic progression occurs in Duverney–Tachiya [1, Section 2, (2.3)–(2.9)]. In the criteria of Kaneko–Suzuki–Tachiya [2, Theorems 1 and 3], sparsity concerns the positions of nonzero power-series coefficients, not the set of selected Lambert denominators. The divisor counts $c_A(n) = \#\{a \in A : a \mid n\}$ are positive on every multiple of $\min A$ for nonempty A , so those criteria do not apply directly. The corresponding density calculation, and the distinction between their remote-tail average and our displacement, are given in [Section 1.2 of the companion paper](#).

The sets of primes for which the weighted sum converges can also be prescribed. Let E be a finite set of primes and let $\mathcal{U}$ be an upward-closed family of subsets of E containing E but not $\emptyset$. A finite union of the constructions just described has divergent reciprocal sum, and, for every $b \geq 2$ and finite prime set P , its weighted sum $W_{b,P}$ is finite exactly when $P \cap E \in \mathcal{U}$. Taking $P = E$ shows that every infinite subset has an irrational sum at every integer base. The construction is given in [Section 1.3 of the companion paper](#).

3 Positive divisor majorants

The weighted criterion treats exponents individually. A divisor cover can take advantage of correlations between several exponents. The relevant quantity is how many members of a finite set divide the same integer.

We shall also use the divisor counts

$$c_A(n) = \#\{a \in A : a \mid n\}, \quad X_A(b) = \sum_{n \geq 1} c_A(n) b^{-n}. \quad (6)$$

Expanding $(b^a - 1)^{-1} = \sum_{j \geq 1} b^{-aj}$ and interchanging nonnegative sums gives the identity. Although $\mathbf{1}_A$ takes only the values zero and one, its divisor transform may take larger values, with $0 \leq c_A(n) \leq \tau(n)$.

3.1 A finite divisor set and the cover condition

For finite $F \subseteq \mathbb{N}_{>0}$, write $f_F(n) = \#\{a \in F : a \mid n\}$. Consider the four exponents $F = \{2, 6, 10, 30\}$. If $2 \nmid n$, the count is zero. Otherwise each of the primes 3 and 5 that divides n doubles the count. Thus

$$f_F(n) = \mathbf{1}_{2|n}(1 + \mathbf{1}_{3|n})(1 + \mathbf{1}_{5|n}).$$

For $0 < \alpha \leq 1$, put $z = 2^\alpha - 1$. Taking the fractional power and expanding gives a positive divisor sum:

$$f_F(n)^\alpha = \mathbf{1}_{2|n} + z\mathbf{1}_{6|n} + z\mathbf{1}_{10|n} + z^2\mathbf{1}_{30|n}, \quad C = \frac{1}{2} \left(1 + \frac{z}{3}\right) \left(1 + \frac{z}{5}\right).$$

Here C is the sum of each coefficient divided by its divisor. These reciprocals are the densities of the divisibility events, so C is also the complete-period mean of this majorant. The same expansion applies to $\{qd : d \mid \prod_{p \in P} p\}$ when no prime in P divides q .

The fractional power replaces each factor 2 in the divisor count by $1 + z$. When the set involves many primes and α is small, this reduces the coefficients in the positive expansion. The tail estimate below also contains the factor $(2^\alpha - 1)^{-1} = 1/z$, which grows as α decreases. The criterion therefore requires summability of the full expression, including this factor, over a sequence of finite sets.

Theorem 3.1 (a summable divisor-cover criterion). *For each $j \geq 1$, let $F_j \subseteq \mathbb{N}_{>0}$ be finite, let $0 < \alpha_j \leq 1$, and let $c_{j,d} \geq 0$ satisfy*

$$f_{F_j}(n)^{\alpha_j} \leq \sum_{d|n} c_{j,d} \quad (n \geq 1).$$

Set $C_j = \sum_{d \geq 1} c_{j,d}/d$. If

$$\sum_{j \geq 1} \frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} < \infty, \quad (V)$$

then $X_A(b)$ is irrational for every infinite $A \subseteq \bigcup_j F_j$ and every integer $b \geq 2$.

For example, take $F_j = \{4^j\}$, $\alpha_j = 1$ and $c_{j,4^j} = 1$, with all other coefficients zero. Then $C_j = 4^{-j}$, and the cost in (V) is $\sum_j 2^{-j}$. A majorant exists for any finite F_j : the choice $\alpha_j = 1$ and $c_{j,d} = \mathbf{1}_{F_j}(d)$ suffices. What restricts an infinite support is summability of the entire sequence of costs. Section 5 gives a necessary condition that applies to every possible cover.

3.2 A finite mean for shifted divisor tails

A divisor majorant becomes a sum of geometric tails, one for each divisor. Indeed, the positive offsets r with $d \mid N + r$ are $d - (N \bmod d), 2d - (N \bmod d), \dots$, so

$$\sum_{\substack{r \geq 1 \\ d \mid N+r}} B^{-r} = \frac{B^{N \bmod d}}{B^d - 1}.$$

Raising a base-two tail to the power α replaces its geometric factor by $B = 2^\alpha$. We therefore need to estimate this expression for $1 < B \leq 2$, including B close to 1. In the mixed theorem, the weighted construction will choose the modulus and dyadic scales after the finite covering sets have been fixed. The estimate must therefore retain these parameters.

For $1 < B \leq 2$, positive integers L, d, M , and an integer $R \geq 0$, put

$$w_{B,d}(n) = \frac{B^{n \bmod d}}{B^d - 1}, \quad \mathcal{D}_{L;R,M} F = \frac{1}{M} \sum_{j=R}^{R+M-1} \frac{1}{2^j} \sum_{m=1}^{2^j} F(Lm).$$

This is a two-stage average: choose the scale j uniformly from $R, \dots, R + M - 1$, then choose m uniformly from $1, \dots, 2^j$ and observe $N = Lm$. Each scale receives equal total weight, although the longer scales contain more observations. The same integer N can occur at several scales, with its contributions added in the displayed sum.

The finite estimate

$$\mathcal{D}_{L;R,M} w_{B,d} \leq \frac{1 + 4L/M}{d(B - 1)} \quad (\text{S})$$

equivalently bounds $(B - 1)\mathcal{D}_{L;R,M} w_{B,d}$ by $(1 + 4L/M)/d$. This normalised bound is uniform as $B \downarrow 1$; the unnormalised right side grows like $(B - 1)^{-1}$. With $B = 2^\alpha$, this is the factor $(2^\alpha - 1)^{-1}$ in (V). The error depends on L/M , independently of the starting scale R . Consequently the bound will still be useful on the weighted proof's windows, provided their number dominates the chosen modulus.

Fix $T = 2^j$. We divide the proof according to how d compares with LT , which is the largest sampled multiple. When $d \leq LT$, the complete orbit has length d/g , with $g = (L, d)$, and total weight $1/(B^g - 1)$. Complete cycles and one remaining piece give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) \leq \frac{g}{d(B^g - 1)} + \frac{1}{T(B^g - 1)} \leq \frac{1}{d(B - 1)} + \frac{1}{T(B - 1)}.$$

Across dyadic lengths satisfying $d \leq L2^j$, the reciprocal-length errors sum to at most $2L/[d(B - 1)]$. When $d > 2LT$, we have $Lm \bmod d = Lm$ and $2Lm \leq d - 1$; hence $\sum_{i=0}^{d-1} B^i \geq dB^{(d-1)/2} \geq dB^{Lm}$. Each term $w_{B,d}(Lm)$ is then at most $1/[d(B - 1)]$. Only the transition range $LT < d \leq 2LT$ remains. For fixed d , doubling T moves from $d > 2LT$ to $d \leq LT$ after at most one intermediate scale, so this range occurs for at most one dyadic length. For that length, the geometric sum and convexity give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) = \frac{B^L}{T(B^L - 1)} \frac{B^{LT} - 1}{B^d - 1} \leq \frac{LB^L}{d(B^L - 1)} \leq \frac{2L}{d(B - 1)}.$$

The first inequality uses the convexity of $x \mapsto B^x - 1$ and its value 0 at the origin to bound the ratio by LT/d . For the last inequality we used $(B^L - 1)/(B - 1) = \sum_{i=0}^{L-1} B^i \geq B^{L-1}$ and $B \leq 2$. Summing the main terms and the two error bounds proves (S). The corresponding [source estimate](#) uses the same finite average. Nonnegative interchange permits summation against any coefficients c_d with $\sum_d c_d/d < \infty$.

3.3 Proof of the cover criterion

Proof of Theorem 3.1. Write $B_j = 2^{\alpha_j}$ and

$$U_j(N) = \sum_{r \geq 1} 2^{-r} f_{F_j}(N+r), \quad V_j(N) = \sum_{d \geq 1} c_{j,d} w_{B_j,d}(N).$$

We first pass from divisor counts to the shifted tails U_j . The power α_j changes their geometric factor to B_j^{-r} , so subadditivity and the assumed majorant yield

$$U_j(N)^{\alpha_j} \leq \sum_{r \geq 1} B_j^{-r} f_{F_j}(N+r)^{\alpha_j} \leq \sum_{d \geq 1} c_{j,d} \sum_{\substack{r \geq 1 \\ d|N+r}} B_j^{-r} = V_j(N).$$

Summing the progression in each residue class gives the last equality. For the displacement itself, the geometric-series identity gives

$$\Delta_{2,F_j}(N) = U_j(N) - X_{F_j}(2) \leq U_j(N).$$

We will make the displacements of the first J finite sets vanish by divisibility, then use $U_j(N)$ to bound the terms from the remaining sets. The early tails themselves need not vanish: the displayed identity then gives $U_j(N) = X_{F_j}(2)$ for $j \leq J$.

Fix $\varepsilon > 0$ and allot $t_j = \varepsilon 2^{-j}$ to the j th tail; these allowances sum to ε . Since $U_j^{\alpha_j} \leq V_j$, it suffices to make $t_j^{-\alpha_j} V_j < 1$. This normalisation explains the factor $2^{j\alpha_j}$ in the cover condition. Choose J with

$$K_J := \sum_{j > J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{4}.$$

This is possible since $\varepsilon^{-\alpha_j} \leq \max(1, \varepsilon^{-1})$. Choose L divisible by every member of the first J finite sets. Equation (S) bounds the finite mean of $S_J(N) := \sum_{j > J} t_j^{-\alpha_j} V_j(N)$ by $(1 + 4L/M)K_J < 1/2$ whenever $M \geq 4L$. We choose a sample with $S_J(N) < 1$. Every summand is nonnegative, so this one choice gives $U_j(N) < t_j$ for all $j > J$ simultaneously. Every exponent in the first J finite sets divides L , so those terms have zero displacement. Consequently,

$$0 < \Delta_{2,A}(N) \leq \sum_{j > J} U_j(N) \leq \varepsilon.$$

Equation (D) now excludes rationality at base two. For $0 \leq r < d$, the function $(b^r - 1)/(b^d - 1)$ is nonincreasing on $b > 1$. For $r > 0$, cancel $b - 1$ and write it as $A(b)/(A(b) + C(b))$, where $A(b) = \sum_{i < r} b^i$ and $C(b) = \sum_{r \leq j < d} b^j$. Its derivative has the sign of $A'C - AC' = \sum_{i < r \leq j < d} (i - j)b^{i+j-1} < 0$. Thus $\Delta_{b,A}(N) \leq \Delta_{2,A}(N)$ for every real $b \geq 2$; when b is an integer, this contradicts the lower bound obtained from (D) under rationality. Any prescribed positive integer can be included as a divisor of L , so the chosen indices can also be required to be arbitrarily large. $\square$

The same proof permits any positive weights η_j with $\sum_j \eta_j = 1$: replace $2^{j\alpha_j}$ in (V) by $\eta_j^{-\alpha_j}$ and take $t_j = \varepsilon \eta_j$. The stronger condition $\sum_j C_j 2^{j\alpha_j} 2^{\alpha_j} / (2^{\alpha_j} - 1)^2 < \infty$ implies (V), since each of its summands is the corresponding summand of (V) multiplied by $2^{\alpha_j} / (2^{\alpha_j} - 1) > 1$. This proves inclusion of the hypotheses for a given cover; after optimisation over all covers, the inclusion of support classes is strict by a [corresponding source theorem](#).

4 A common index for the two criteria

The cover estimate retains its dependence on the modulus so that it can use the larger modulus chosen by the weighted construction. This lets us add the two nonnegative errors on one finite distribution and only then choose an index. We work at base two and obtain the all-base conclusion by the displacement comparison proved above.

Theorem 4.1 (mixed weighted and cover supports). *Let $E, V \subseteq \mathbb{N}_{>0}$. Suppose E satisfies (W) for a finite nonempty prime set P , and $V \subseteq \bigcup_j F_j$ for finite sets and nonnegative majorants satisfying the hypotheses of Theorem 3.1, with either (V) or its positive-weight variant. Then $X_A(b)$ is irrational for every infinite $A \subseteq E \cup V$ and every integer $b \geq 2$.*

Proof. Fix $\varepsilon > 0$ and an integer $N_0 \geq 1$, and put $\rho = \varepsilon/3$. Use the cover notation B_j, U_j, V_j above, with $\eta_j = 2^{-j}$ in the case (V). Set $t_j = \rho \eta_j$ and choose J so that

$$K_J = \sum_{j>J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{16}.$$

Choose a finite $F \subseteq E$ so that the weighted sum over $E \setminus F$ is $\kappa < \rho/16$. Let L be a positive common multiple of N_0 , all members of F , and all members of the first J finite sets. The order of these choices matters: first J, F and hence L are fixed; then the weighted argument enlarges L to a modulus Q and chooses its dyadic scales; the sample is selected only after both estimates have been placed on that distribution. With J, F and L fixed, use the base-two choices

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \lfloor H / \max P \rfloor, \quad M = \lfloor 2^{G/2} \rfloor.$$

Both estimates below use $\mathcal{D}_{Q;M,M}$: the scale runs from M to $2M-1$, and each observation is a multiple $N = Qm$. We keep this average fixed while estimating the two contributions. The proof of (5) allows any multiple of F as L , including the extra divisors imposed by the cover. The estimate also holds for finite or empty E , since infinitude was used only to make the final displacement positive. Consequently

$$\mathcal{D}_{Q;M,M}(\Delta_{2,E}/\rho) < \frac{1}{8}$$

for sufficiently large H . For the same finite distribution, (S) and nonnegative interchange give

$$\mathcal{D}_{Q;M,M} S_J \leq (1 + 4Q/M) K_J < \frac{1}{8}, \quad S_J = \sum_{j>J} t_j^{-\alpha_j} V_j,$$

because $Q/M \rightarrow 0$. The dependence on the modulus in (S) is essential here: the covering sets fixed L , whereas the actual samples use the larger modulus Q . Adding on this one distribution gives

$$\mathcal{D}_{Q;M,M}(\Delta_{2,E}/\rho + S_J) < \frac{1}{4}.$$

Choose a sampled $N = Qm$ for which the nonnegative sum is less than 1. Then $\Delta_{2,E}(N) < \rho$, and each summand of $S_J(N)$ is less than 1. Since $U_j^{\alpha_j} \leq V_j$, the same choice gives $U_j(N) < t_j$ for every $j > J$. The displacement terms from the first J finite sets vanish because their exponents divide L , and hence divide N . Therefore

$$\Delta_{2,A}(N) \leq \Delta_{2,E}(N) + \Delta_{2,V}(N) < 2\rho < \varepsilon, \quad N \geq Q \geq L \geq N_0.$$

This inequality remains valid when E and V overlap. Since A is infinite, $\Delta_{2,A}(N) > 0$. The comparison $0 < \Delta_{b,A}(N) \leq \Delta_{2,A}(N)$ proved in the cover argument therefore gives arbitrarily small positive values at every integer base. Equation (D) rules out rationality. $\square$

5 Comparison of the support classes

Write $\mathcal{W}_b$ for supports satisfying the finite-prime weighted hypothesis at base b , and $\mathcal{C}$ for supports admitting a divisor cover satisfying (V). Neither condition includes the other.

Proposition 5.1 (incomparable support criteria). *There are infinite positive supports E and V such that*

$$E \in \mathcal{W}_2, \quad E \notin \mathcal{C}, \quad V \in \mathcal{C}, \quad V \notin \mathcal{W}_b \ (b \geq 2), \quad \sum_{a \in V} \frac{1}{a} = \infty.$$

Both constructions use finite sets of the form $\{qd : d \mid \prod_{p \in P} p\}$. For E , we take q to be a growing power of 2. This makes the weighted sum converge, while the many simultaneous divisors force the lower bound for any cover to diverge. For V , we use disjoint sets of primes in successive finite sets. For any fixed finite set of primes, all sufficiently late sets contain only exponents coprime to those primes, so the weighted sum diverges. Fractional powers of the divisor counts will nevertheless give a summable cover cost.

5.1 A lower bound independent of the cover

To prove that E has no cover, it is not enough to make one choice of majorants expensive. We need a lower bound depending only on finite subsets of E . For finite F , let $\mathbb{E}_F$ denote the uniform mean modulo $\text{lcm}(F)$, with $\text{lcm}(\emptyset) = 1$ and $f_\emptyset = 0$. Allow arbitrary weights $\eta_j > 0$ with $\sum_j \eta_j = 1$, so that the lower bound also survives reweighting the cover. Set

$$K = \sum_j \frac{C_j \eta_j^{-\alpha_j}}{2^{\alpha_j} - 1}, \quad \Psi(0) = 0, \quad \Psi(t) = \inf_{0 < \alpha \leq 1} \frac{t^\alpha}{2^\alpha - 1} \quad (t \geq 1).$$

Here $\log^+ t = \log \max\{1, t\}$. For every finite F contained in $\bigcup_j F_j$,

$$K \geq \mathbb{E}_F \Psi(f_F) \geq e \mathbb{E}_F \log^+ f_F. \quad (7)$$

Indeed, if $f_F(n) = t > 0$, coverage gives $\sum_j f_{F_j}(n) \geq t$, so some j satisfies $f_{F_j}(n) \geq \eta_j t$. Its weighted majorant is at least $t^{\alpha_j} / (2^{\alpha_j} - 1) \geq \Psi(t)$. The index j may depend on n : it is the sum of all weighted majorants that bounds $\Psi(f_F(n))$ at every n . Average this pointwise inequality over $1 \leq n \leq X$. Each divisor indicator has mean $\lfloor X/d \rfloor / X \leq 1/d$, so the mean of the sum of majorants is at most K . As $X \rightarrow \infty$, the mean of $\Psi(f_F)$ tends to $\mathbb{E}_F \Psi(f_F)$. Only this finite-support function needs a period; the covering moduli need not divide $\text{lcm}(F)$. This proves the first inequality. The second is immediate for $t = 0, 1$; for $t > 1$, use $2^\alpha - 1 \leq \alpha$ and $e^u / u \geq e$ with $u = \alpha \log t$. The scalar inequality $t^\alpha / (2^\alpha - 1) \geq e \log t$ also has a [corresponding source theorem](#).

The lower bound is asymptotically attained on the sets $\{qd : d \mid \prod_{p \in P} p\}$: (13) in Appendix E gives the sharp leading constant. The separation below uses only the lower bound (7).

5.2 Two separating constructions

Proof of Proposition 5.1. The reciprocal sum over odd primes diverges, even after finitely many are removed. Otherwise $\prod_{p \text{ odd}} (1 + 1/p)$ would bound the sum of reciprocals of odd squarefree integers; the decomposition $n = ds^2$ would then make the odd harmonic series converge.

For the first construction, the k th group of exponents will carry a factor 2^k . The event $v_2(n) = k$ has density $2^{-(k+1)}$, so a prime reciprocal sum of order 2^k will give a fixed contribution to the mean logarithmic count. Taking its leading constant below $\log 2$ leaves exponential decay in the weighted sum. Choose pairwise disjoint finite sets P_k of odd primes such that $2^k/4 \leq S_k := \sum_{p \in P_k} 1/p < 2^k/4 + 1$. Put $M_k = \prod_{p \in P_k} p$ and $E = \bigcup_{k \geq 1} \{2^k d : d \mid M_k\}$. For $P = \{2\}$ the weighted sum is

$$\sum_{k \geq 1} \frac{\prod_{p \in P_k} (1 + 1/p)}{2^{2^k} - 1} \leq 2e \sum_{k \geq 1} \exp(-(\log 2 - 1/4)2^k) < \infty.$$

For $F_m = \bigcup_{k=1}^m \{2^k d : d \mid M_k\}$, average $\log^+ f_{F_m}$ over one period. On $v_2(n) = k < m$, the set $\{2^k d : d \mid M_k\}$ contributes $2^{Z_k(n)}$ divisors, where $Z_k(n) = \sum_{p \in P_k} \mathbf{1}_{p|n}$. This event has density $2^{-(k+1)}$ and is independent of odd-prime divisibility. Thus

$$\mathbb{E}_{F_m} \log^+ f_{F_m} \geq (\log 2) \sum_{k=1}^{m-1} 2^{-(k+1)} S_k \geq \frac{m-1}{8} \log 2.$$

Bound (7) makes these means uniformly bounded for every strengthened cover, so $E \notin \mathcal{C}$.

For the reverse direction, choose unused odd primes $q_j \geq 4^j$ and disjoint finite sets P_j of unused odd primes. Stop each P_j at the first prefix with $R_j := \prod_{p \in P_j} (1 + 1/p) \geq q_j$. Prime reciprocal divergence permits this, and minimality gives $q_j \leq R_j < 4q_j/3$. Put

$S_j = \sum_{p \in P_j} 1/p$, $M_j = \prod_{p \in P_j} p$, and $V = \bigcup_{j \geq 1} F_j$, where $F_j = \{q_j d : d \mid M_j\}$. No prime divides exponents in two different sets F_j , and

$$\sum_{a \in F_j} \frac{1}{a} = \frac{R_j}{q_j} \in [1, 4/3).$$

Thus $\sum_{a \in V} 1/a$ diverges. Given a finite set P of primes, all exponents in all but finitely many F_j are coprime to $\prod_{p \in P} p$. On each such F_j we have $h_P(a) = 1$, so its contribution to the base- b weighted sum is $R_j/[q_j(b-1)] \geq 1/(b-1)$. Hence $V \notin \mathcal{W}_b$ for every integer $b \geq 2$.

For the full infinite-cover cost, from $\log(1+x) \geq x - x^2/2$ and $\sum_{p \in P_j} p^{-2} \leq 1$ we have $S_j \leq \log R_j + 1/2 < \log q_j + 1$. In the j th term of the cover cost, the factor $2^{j\alpha_j}$ becomes $(1+z_j)^j$, and the divisor sum contributes $\prod_{p \in P_j} (1+z_j/p)$. Their product is at most $\exp(z_j(j+S_j))$, which we keep bounded by choosing $z_j(j+S_j) = 1$. Set

$$z_j = (j+S_j)^{-1}, \quad \alpha_j = \log_2(1+z_j), \quad c_{j,q_j d} = z_j^{\omega(d)} \quad (d \mid M_j),$$

with other coefficients zero and $\omega(d)$ the number of prime factors of d . If $q_j \mid n$ and Z primes in P_j divide n , the divisor majorant equals $(1+z_j)^Z = f_{F_j}(n)^{\alpha_j}$; otherwise both sides vanish. The sum defining C_j is $C_j = q_j^{-1} \prod_{p \in P_j} (1+z_j/p)$, whence

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} = \frac{(1+z_j)^j}{q_j z_j} \prod_{p \in P_j} (1+z_j/p) \leq \frac{e(j+S_j)}{q_j} \leq e((1+\log 4)j+1)4^{-j}.$$

The last bound uses that $(j+\log x+1)/x$ decreases for $x \geq 4^j$. The bound is summable in j and includes the factor $2^{j\alpha_j}$ required by (V); hence $V \in \mathcal{C}$. $\square$

5.3 A strict enlargement and finite unions

The union of the two separating supports requires the mixed theorem.

Corollary 5.2 (a support requiring the mixed criterion). *There is an infinite positive support U with $U \notin \mathcal{C}$ and $U \notin \mathcal{W}_b$ for every integer $b \geq 2$, such that $X_A(b)$ is irrational for every infinite $A \subseteq U$ and every integer $b \geq 2$.*

Proof. Let $U = E \cup V$, with E, V as in Proposition 5.1. A cover of U is also a cover of E , while convergence of the weighted sum over U would imply convergence of the sum over V for the same prime set, since all summands are nonnegative. Thus U belongs to neither individual class. Theorem 4.1 gives irrationality for every infinite subset of U at every integer base $b \geq 2$. $\square$

The fixed-dyadic cover class in (V) is closed under finite unions. When interleaving two covers, we must control the index-dependent factor $2^{j\alpha_j}$. Place their j th finite sets at positions $2j-1, 2j$ and halve each exponent; this factor then does not increase. The old majorant coefficients still work, since $f^{\alpha/2} \leq f^\alpha$ for every nonnegative integer f . It

remains to compare the denominators in the cost. Writing $T_j = C_j 2^{j\alpha_j} / (2^{\alpha_j} - 1)$ for the original j th cost, its new cost is at most

$$\frac{C_j 2^{j\alpha_j}}{2^{\alpha_j/2} - 1} = (1 + 2^{\alpha_j/2}) T_j \leq (1 + \sqrt{2}) T_j.$$

For r covers, interleave at positions at most rj and divide each exponent by r ; the cost ratio is $(2^\alpha - 1)/(2^{\alpha/r} - 1) < 2r$. Weighted supports are likewise closed under finite unions: take the union of their finite prime sets. The prime part $h_p(a)$ can only increase, and $h/(b^h - 1)$ decreases with h . Finite supports satisfy both criteria, and each criterion is inherited by subsets. Thus the mixed class is closed under finite unions and finite changes with the fixed dyadic weights in (V); arbitrary cover weights are not needed. The summability hypothesis cannot be omitted for countable unions. Every prime singleton is admitted, but for the full prime support $\mathcal{P}$ one has $\Delta_{2,\mathcal{P}}(N) > 1/3$ for every $N \geq 1$: indeed $\sum_{r \geq 1} 2^{-r} \omega(N+r) \geq 1$, whereas $X_{\mathcal{P}}(2) \leq \sum_{a \geq 2} (2^a - 1)^{-1} < 2/3$. Here $\omega(n)$ is the number of distinct prime divisors of n . The reciprocal-summable class is contained in the weighted class, since $h/(2^h - 1) \leq 1$. A direct averaging proof of this special case is given in Section A.

For comparison, Tao–Teräväinen prove the full-prime case at base 2 [3, Theorem 1.3, p. 4]. The paragraph following it states extensions to prime support at every integer base and to full prime-power support at base 2, leaving the modifications to the reader. It does not treat arbitrary infinite thinnings. The proposed thinning extension is not a premise of any theorem here.

6 Limits of the small-displacement criterion

The small-displacement condition does not cover even every support whose sum is already known to be irrational. For full support we have

$$\Delta_{2,\mathbb{N}_{>0}}(N) > (2^N - 1) \sum_{a > N} 2^{-a} = 1 - 2^{-N} \geq \frac{1}{2} \quad (N \geq 1).$$

The full-support value is nevertheless [irrational](#) [5]. Thus a proof for arbitrary support would have to use more than this particular family of small displacements. The example leaves other integer linear forms and averaging arguments available.

The logarithmic lower bound is not a converse: replacing a fractional majorant by a logarithmic one need not control averages along the multiples of a prescribed modulus. The counterexample and the bound that remains valid on ordinary initial intervals are proved in [Section 13 of the companion paper](#); those proofs were first worked out in an AI-assisted note of 17 September 2026. These are statements about finite averages; they do not rule out an irrationality criterion based on other uses of logarithmic divisor counts.

For squarefree support, Duverney and Tachiya’s theorem already gives irrationality. It holds at every base 2^j , $j \geq 1$, by Corollary 1.2 and Example 1.1 of Duverney and Tachiya [1, p. 4]. Their corollary covers the s -free products of any pairwise coprime sequence of polynomial growth, and they present it as support for the conjecture of Erdős

and Graham stated here as Problem 1.1. For the squarefree support with 1 removed, the divisor count is $2^{\omega(n)} - 1$, which is odd for $n \geq 2$. This excludes a first-block-divisibility condition at even bases. Adjoining 1 removes that parity obstruction while changing the sum by a rational number. The argument concerns that normalisation of the certificate, and gives no obstruction to irrationality of the sum. Details and the other known support families appear in [the comparison with known support theorems in the companion paper](#).

A growth-only criterion gives a different class: if $A = \{c_1 < c_2 < \dots\}$ and $\limsup_n c_n/2^n = \infty$, then $X_A(b)$ is irrational for every integer $b \geq 2$ [8, Theorem 1]. The same section of the companion paper proves the translation from denominator growth and gives examples showing that this class and reciprocal summability are incomparable. The interval-filling constructions in [9] and the freely chosen lacunary denominators of [16] are not constructions of subseries of these fixed Mersenne weights.

Prime-incidence arguments require additional care as well: multiplying all prime exponents by 2 creates positive correlations between their divisibility indicators. The identity and covariance calculation appear at the end of [the companion paper's discussion of mixed supports](#). Size conditions alone do not supply that independence.

6.1 Algebraic bases on separated supports

The companion [Reading eight Erdős problems together](#), subsection “Divisibility cuts at every algebraic base”, states a transcendence extension based on the number-field Subspace Theorem. Its hypothesis is an infinite set H with indices $L_j < M_j$, where $L_j \rightarrow \infty$ and $M_j - L_j \rightarrow \infty$, such that every $n \in H$ with $n \leq L_j$ divides L_j , and every $n \in H$ with $n > L_j$ is a multiple of M_j . The claimed conclusion is transcendence of $\sum_{n \in B} w_n/(t^n - 1)$ for every infinite $B \subseteq H$, bounded positive integer weights w_n and real algebraic $t > 1$.

Every divisibility chain has such cuts, as does $H_* = \bigcup_j N_j \{1, \dots, 2^{N_j}\}$ with $N_0 = 1$ and $N_{j+1} = 2N_j \text{ lcm}(1, \dots, 2^{N_j})$. Each block has a reciprocal sum of at least $1/2$ and contains antichains of unbounded size. The weighted sum for one prime is at most $2t^2/(t-1)^3$ at every real $t > 1$. This connects the example with the weighted criterion, although the separation condition is not asserted for every support satisfying that criterion.

The companion proof is outside this paper and is not a premise of the weighted, cover or mixed theorem. It has two AI proof reviews, no Lean proof of transcendence and no independent human review. No historical priority is asserted for the extension. The case of arbitrary infinite support at base two remains unresolved.

A Reciprocal-summable supports

Erdős stated the following extension of his pairwise-coprime theorem [7, p. 222]. It follows from Theorem 1.2, since $h/(2^h - 1) \leq 1$, but the direct argument explains why reciprocal summability makes the averaging simpler.

Theorem A.1 (reciprocal-summable supports). *Let $A \subseteq \mathbb{N}_{>0}$ be infinite. If*

$$\sum_{a \in A} \frac{1}{a} < \infty,$$

then $X_A(b)$ is irrational for every integer $b \geq 2$.

Erdős also discussed weaker conditions [7, pp. 222, 226]. We do not identify our weighted hypothesis with the conditions he suggested.

Here we can take the observation length to infinity while holding the modulus fixed: reciprocal summability supplies a summable bound for all the remaining exponents. We then enlarge the modulus to make each fixed exponent divide it. No common period for the whole infinite support is required.

Fix the integer base $b \geq 2$ and put

$$w_{b,d}(N) = \frac{b^{N \bmod d}}{b^d - 1}, \quad T_N^{(b)} = \sum_{d \in A} w_{b,d}(N).$$

Then $T_N^{(b)} - T_0^{(b)} = \Delta_{b,A}(N)$. For a fixed positive integer Q , the orbit of Q modulo d consists of d/g residues, where $g = (Q, d)$. Summing the geometric progression gives

$$M_{Q,b}(d) := \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) = \frac{g}{d(b^g - 1)} \leq \frac{1}{d}. \quad (8)$$

The orbit $4, 2, 0$ modulo 6 used in Section 2 illustrates this mean: its average of $w_{2,6}(4m)$ is $1/9$, while the average of $w_{2,6}(4m) - w_{2,6}(0)$ is $2/21$. When $d \mid Q$, we have $w_{b,d}(Qm) = w_{b,d}(0)$, so this exponent contributes zero to the displacement.

To interchange the sum over exponents and this limit, we use a bound summable in d for the fixed modulus Q . Counting positive multiples of d gives

$$\begin{aligned} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) &= \sum_{r \geq 1} b^{-r} \frac{\#\{1 \leq m \leq X : d \mid Qm + r\}}{X} \\ &\leq \sum_{r \geq 1} b^{-r} \frac{Q+r}{d} = \frac{Q/(b-1) + b/(b-1)^2}{d} \leq \frac{Q+2}{d}. \end{aligned}$$

Indeed, the counted multiples are distinct and at most $QX + r$. For this fixed Q , the majorant $(Q+2)/d$ is summable over A . Dominated convergence therefore takes the observation-length limit inside the sum over d , giving the Cesàro mean $\sum_{d \in A} M_{Q,b}(d)$. This majorant is not uniform in growing Q .

Next let $Q_t = \text{lcm}(1, \dots, t)$. For each fixed d , eventually $d \mid Q_t$ and $M_{Q_t,b}(d) = (b^d - 1)^{-1}$. A second dominated-convergence argument, using $M_{Q_t,b}(d) \leq 1/d$, yields

$$\sum_{d \in A} M_{Q_t,b}(d) \rightarrow X_A(b). \quad (9)$$

Thus the limiting averages of the nonnegative displacements $\Delta_{b,A}(Q_t m)$ tend to zero. Choose t , then a sufficiently long finite average, and finally a term no larger than that average. This gives arbitrarily small positive displacements, contradicting the lower bound $1/q$ obtained from (D). This proves Theorem A.1 directly at every integer base.

The order of limits is essential: the observation length tends to infinity with the modulus fixed, and only then does the modulus increase. Reciprocal summability controls both interchanges.

For example, every powerful integer (each prime factor occurs to exponent at least two) has the form u^2v^3 . Hence its reciprocal sum is at most $\zeta(2)\zeta(3)$, and the theorem applies to every infinite subset of the powerful integers. For full perfect-power supports, compare Duverney–Tachiya [1, Corollary 1.2, p. 4] and the earlier linear-independence results of Luca–Tachiya [13].

B Finite-support denominator periods

For a finite nonempty $F \subseteq \mathbb{N}_{>0}$, let D_F be the positive reduced denominator of $X_F(b) = \sum_{n \in F} (b^n - 1)^{-1}$. We use $\text{ord}_1(b) = 1$.

Theorem B.1 (the exact denominator period). *Let $F \subseteq \mathbb{N}_{>0}$ be finite and nonempty, let $b \geq 2$ be an integer, and let $D_F > 0$ be the denominator of $X_F(b)$ in lowest terms. Then D_F is coprime to b , and*

$$\text{ord}_{D_F}(b) = \text{lcm}\{n : n \in F\}.$$

If moreover $\text{lcm}(F) \geq 2$, then $\text{lcm}(F) < D_F$. We use $\text{ord}_1(b) = 1$, so the statement includes $F = \{1\}$ at $b = 2$.

Put $L = \text{lcm}(F)$. Clearing denominators gives $D_F \mid b^L - 1$, so the upper divisibility for the order is immediate. For the reverse, choose $n \geq 2$ maximal under divisibility in F and a prime $\ell \mid \Phi_n(b)$. If $e = v_\ell(b^n - 1)$, the full prime power ℓ^e has $\text{ord}_{\ell^e}(b) = n$. Every other selected exponent m has $n \nmid m$, hence $v_\ell(b^m - 1) < e$. The n th summand has uniquely smallest ℓ -adic valuation and cannot cancel. Thus $\ell^e \mid D_F$ and $n \mid \text{ord}_{D_F}(b)$. Taking all maximal selected exponents proves the order statement; the size bound follows from $\text{ord}_{D_F}(b) \mid \varphi(D_F) < D_F$ when $L \geq 2$. The case $F = \{1\}$ has order one directly.

The unique-valuation argument also permits signs ± 1 on the finite summands. This signed extension is not included in the formal statement. The required cyclotomic prime-power fact is proved in Section E. The distinction between primes and prime powers is visible in

$$X_{\{2,3\}}(2) = \frac{10}{21}, \quad X_{\{2,6\}}(2) = \frac{22}{63}.$$

In both examples 2 has multiplicative order six modulo the denominator. The first combines orders two and three; the second retains the order-six prime power 9, although no prime divisor of 63 has order six. The boundary $F = \{1\}$ at base two has $D_F = L = 1$. These lower denominator bounds give no upper height control for infinite partial sums and do not decide an infinite-support value. Van Assche instead constructs approximants for the full Lambert series and proves both nonvanishing and decay of the associated integer linear forms [14, Lemma 1 and (34)–(35)]. That approximation mechanism does not follow from denominator survival alone.

In particular, $1/21$ is not a finite subseries sum. If it were, the order of 2 modulo 21 would force every selected exponent to divide 6. The exponent 1 is excluded since its weight is 1. Any sum containing 2 or 3 exceeds $1/21$, whereas the only nonzero remaining sum is $1/63$, from the exponent 6. This will distinguish finite and infinite representations in the next appendix.

C Rational membership

We now fix a target value rather than impose a condition on its support. The first recurrence is forced by rationality; the later greedy recurrence tests whether a specified target is represented.

C.1 Rational values and scaled tails

Suppose $X_A(2) = p/v$, where $p \in \mathbb{Z}$ and $v \geq 1$ is an integer. Multiplying the coefficient-series identity by $v2^N$ gives

$$z_N := v \sum_{r \geq 1} c_A(N+r)2^{-r} = 2^N p - v \sum_{n=1}^N c_A(n)2^{N-n} \in \mathbb{Z}.$$

Separating the first term of the tail then gives $z_{N+1} = 2z_N - vc_A(N+1)$. All the coefficients $c_A(n)$ are divisor counts of the same set A .

Two elementary tail properties hold regardless of whether the value is rational. Assume first that A is nonempty and put $a_0 = \min A$. Then every a_0 consecutive integers contain a multiple of a_0 , so a run of zeros of c_A has length at most $a_0 - 1$. If A is infinite, choose k exponents and a common multiple L ; then $c_A(L) \geq k$ and $\sum_{r \geq 1} c_A(L-1+r)2^{-r} \geq k/2$, by its first term. Both facts hold without rationality. A contradiction must use the integral recurrence together with the requirement that every coefficient counts divisors in the same set A .

The compatibility condition on the coefficients is expressed by Dirichlet convolution: $\mu * c_A = \mathbf{1}_A$, where μ is the Möbius function. A putative recurrence with integral forcing must therefore satisfy $(\mu * c_A)(n) \in \{0, 1\}$ for all n . The telescoping proof of the integer-recurrence criterion and the Möbius inversion argument are given under [Bounds for a general coefficient sequence](#) in the companion paper. The example $A = \{2\}$ there has value $1/3$. The correspondence permits finite supports; infinitude remains a separate requirement for a counterexample to Problem 1.1.

C.2 The achievement set

Let $w_n = (2^n - 1)^{-1}$ and

$$\mathcal{A} = \left\{ \sum_{n \geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0, 1\} \right\}, \quad R_N = \sum_{n > N} w_n.$$

The estimate

$$2^{-N} < R_N \leq 2^{-N} + \frac{2}{3} 4^{-N} < w_N \quad (N \geq 1)$$

implies uniqueness of the sequence (ε_n) . Indeed, at the first differing exponent n , the difference w_n exceeds everything later terms can cancel. The middle inequality follows by writing $w_n = 2^{-n} + 4^{-n}/(1 - 2^{-n})$ and using $n \geq N+1 \geq 2$. Since every weight exceeds its tail, Hornich's theorem [15], as proved by Nitecki [10, Theorem 4(1), p. 9], shows that the coding image is a Cantor set of measure $\lim_N 2^N R_N$; the estimate above gives $2^N R_N \rightarrow 1$, so $\lambda(\mathcal{A}) = 1$. Concretely, fixing the first N digits gives 2^N disjoint closed intervals of length R_N . These interval unions decrease to $\mathcal{A}$, and continuity of

Lebesgue measure from above gives the same limit. Kovač–Tao record this strict-tail inequality and the Cantor conclusion in the fixed-base setting [4, Remark 4.1]. Thus each represented target determines a unique set of selected exponents. The [geometry section of the companion paper](#) also treats restricted supports. For any set J of allowed exponents, each retained weight exceeds its restricted tail, so the digit map is injective. If J omits exactly the finite set F , its achievement set has measure $2^{-|F|}$; if J omits infinitely many exponents, that measure is zero. The [tail comparison](#), [injectivity](#) and [measure dichotomy](#) have separate source statements.

C.3 The greedy recurrence

For $x \geq 0$, the greedy rule starts with $r_0 = x$ and, at rank $n \geq 1$, selects n when $r_{n-1} \geq (2^n - 1)^{-1}$, subtracting that weight if selected. Let A_x be the resulting support and $c_x(n) = \#\{a \in A_x : a \mid n\}$. Set

$$P_0 = 0, \quad P_{N+1} = 2P_N + c_x(N+1), \quad Q_N = \lfloor 2^N x \rfloor - P_N.$$

Thus $P_N = \sum_{j=1}^N 2^{N-j} c_x(j)$ is the integer truncation of the Lambert expansion (6). It is at most $2^N X_{A_x}(2) \leq 2^N x$, so $Q_N \geq 0$. These integer remainders obey

$$Q_{N+1} = 2Q_N + \beta_N - c_x(N+1), \quad \beta_N = \lfloor 2^{N+1} x \rfloor - 2\lfloor 2^N x \rfloor \in \{0, 1\}. \quad (10)$$

The reason to test for nonincreases is a difference in growth rates. For a represented target, Q_N is bounded by a scaled coefficient tail of order $\sqrt{N}$; integrality prevents strict increase throughout a sufficiently long window. For an unrepresented target, the positive deficit $x - X_{A_x}(2)$ contributes exponential growth, and the recurrence eventually forces strict increase.

Theorem C.1 (membership and nonincreasing integer remainders). *For every real $x \geq 0$, the following are equivalent:*

$$\begin{aligned} & x \in A; \\ & \forall K \geq 0 \exists N \geq K : Q_{N+1} \leq Q_N; \\ & \forall K \geq 0 \exists N \in [K, K + 2\lfloor \sqrt{K} \rfloor + 12) : Q_{N+1} \leq Q_N. \end{aligned}$$

Proof. Strict domination of each Mersenne weight over its tail implies that a represented target is recovered by the greedy rule. For such a target,

$$0 \leq Q_N \leq \sum_{r \geq 1} c_x(N+r) 2^{-r} \leq 2\sqrt{N} + 4,$$

since $c_x(n) \leq \tau(n) \leq 2\sqrt{n}$ and $\sqrt{N+r} \leq \sqrt{N} + \sqrt{r} \leq \sqrt{N} + r$. Put $s = \lfloor \sqrt{K} \rfloor$ and $T = 2s + 12$. Strict increase at all T steps would imply $Q_{K+T} \geq T$. But $K+T < (s+4)^2$ gives $Q_{K+T} < 2s + 12 = T$. This proves the window condition and hence cofinal nonincreases.

Conversely, if $x \notin A$, then $\delta = x - X_{A_x}(2) > 0$. The Lambert prefix is at most $2^N X_{A_x}(2)$, so $Q_N \geq 2^N \delta - 1$. Eventually this exceeds $c_x(N+1) \leq N+1$. Equation (10) then makes Q_N strictly increasing, contradicting cofinal nonincreases. $\square$

The nonincreases must occur along the target's own greedy sequence. For $x = 0$, no exponent is selected and $Q_N = 0$ at every rank. A finite subseries sum is likewise represented and satisfies the criterion. The target $x = 3/4$ lies strictly between $R_1 < 2/3$ and $w_1 = 1$, so it is not represented and its integer remainders eventually increase strictly. For $1/2$ and $1/21$, the theorem does not establish which alternative occurs.

The same argument allows a uniform subpower window: for each $0 < \varepsilon < 1$, a constant C_ε gives the window length $\lceil C_\varepsilon(K+1)^\varepsilon \rceil$. Indeed, $\tau(n) \leq A_\varepsilon n^\varepsilon$ gives $Q_N \leq D_\varepsilon(N+1)^\varepsilon$ for represented targets, where $D_\varepsilon = A_\varepsilon \sum_{r \geq 1} r^\varepsilon 2^{-r}$. Choose $C_\varepsilon > D_\varepsilon(C_\varepsilon+3)^\varepsilon$, which is possible because $\varepsilon < 1$. For $T = \lceil C_\varepsilon(K+1)^\varepsilon \rceil$, strict increase at all T steps would give

$$T \leq Q_{K+T} \leq D_\varepsilon(K+T+1)^\varepsilon \leq D_\varepsilon(C_\varepsilon+3)^\varepsilon(K+1)^\varepsilon < T,$$

a contradiction. The converse still follows from exponential growth when x is not represented. This shortens the window for represented targets without deciding whether a specified target is represented. The linked formal proof establishes the square-root window.

At $x = 1/2$, the digits β_N vanish for $N \geq 1$; at $x = 1/21$ they are six-periodic. Thus the unresolved arithmetic input is

$$\forall K \exists N \geq K : \quad c_x(N+1) \geq Q_N + \beta_N, \quad x \in \{1/2, 1/21\}. \quad (11)$$

Both Q_N and c_x must arise from the same greedy support A_x . Counterexamples to fixed-multiplier schedules, with their finite phase masks, are retained in the companion paper. They concern those schedules; they do not decide either target.

D Integer quotients, approximation and further questions

For $1/21$ an integer greedy procedure gives finite remainders that can be compared with the real greedy sequence. For $1/2$ we instead allow arbitrary finite approximating supports. In both cases the outstanding question is whether suitable finite data occur at unbounded depths.

Integer quotients for $1/21$. For $M, d \geq 1$, set

$$q_M(d) = \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor, \quad T_M = \left\lfloor \frac{2^M}{21} \right\rfloor.$$

Starting with the integer remainder T_{2R} , consider $d = 2, \dots, R$ in order and subtract $q_{2R}(d)$ whenever it does not exceed the current remainder. Let D_R be the set of selected exponents and s_R the final remainder. Thus

$$s_R = T_{2R} - \sum_{d \in D_R} q_{2R}(d) \geq 0.$$

A second comparison uses the same integer rule through $d = 2R$, rather than stopping at R . For example, $R = 6$ gives $T_{12} = 195$, $D_6 = \{5\}$ and $s_6 = 195 - \lfloor 4096/31 \rfloor = 63 \leq 64$. This is one row satisfying the bound below, not evidence that such rows occur at unbounded depths.

Write $r_n = r_n(1/21)$ for the real greedy remainder and $A_{1/21}$ for its support. The condition $\mathcal{F}_{21}$ means that there exist integers $n, R_0, K_0 \geq 0$ with all of the following properties:

1. $r_n > R_n$, the set of positive exponents outside $A_{1/21}$ is finite, and every exponent greater than n belongs to $A_{1/21}$;
2. for every $R \geq R_0$, the integer rule with target T_{2R} and weights $q_{2R}(2), \dots, q_{2R}(2R)$ selects exactly the same exponents as the real greedy rule on $\{2, \dots, 2R\}$;
3. for every $K \geq K_0$, the interval $(K, 2K]$ contains an exponent of $A_{1/21}$.

The eventual all-selected behaviour makes some clauses redundant. We retain them to specify both the real greedy sequence and the comparison with its finite integer versions.

Theorem D.1 (integer-quotient tests for $1/21$). *The following statements hold.*

1. $1/21 \in \mathcal{A}$ if and only if $\mathcal{F}_{21}$ does not hold.
2. If there is an unbounded sequence of ranks R with $s_R \leq 2^R$, then $1/21 \in \mathcal{A}$.
3. On $\mathcal{F}_{21}$, eventually $s_R > 2^R$, the boundary rank $R + 1$ belongs to D_{R+1} , and, for all sufficiently large R ,

$$\begin{aligned} D_{R+1} &= D_R \cup \{R + 1\}, \\ s_{R+1} &= 4s_R + \left\lfloor \frac{4(2^{2R} \bmod 21)}{21} \right\rfloor - 2c_{D_R}(2R + 1) - c_{D_R}(2R + 2) \\ &\quad - (2^{R+1} + 1). \end{aligned}$$

Here $c_{D_R}(m) = \#\{d \in D_R : d \mid m\}$.

Proof. Write $x = 1/21$, $G = A_{1/21}$ and $\delta = x - X_G(2) \geq 0$. If $x \notin \mathcal{A}$, then $\delta > 0$. The real remainder tends to δ , so every sufficiently small weight is selected and eventually $r_n > R_n$. For the integer comparisons, normalise by 4^R . As long as the selected exponents agree through rank $d - 1$, the remainders differ by at most $d/4^R$. Each fixed comparison is strict, since equality would give a finite representation of x . Choose d_0 with $w_d < \delta/2$ for $d > d_0$. For large R the comparisons through d_0 agree and $2R/4^R < \delta/2$, so induction forces both rules to select all remaining ranks through $2R$. This proves $\mathcal{F}_{21}$. A represented target has $r_n \leq R_n$ at every rank, giving the converse in (1).

For every R , the rounding errors satisfy

$$\left| x - X_{D_R}(2) - \frac{s_R}{4^R} \right| \leq \frac{R + 1}{4^R}. \quad (12)$$

Hence an unbounded sequence with $s_R \leq 2^R$ gives finite subseries values tending to x . Closedness of $\mathcal{A}$ proves (2), and Section B excludes a finite representation.

On $\mathcal{F}_{21}$, the eventual agreement gives $D_R = G \cap \{2, \dots, R\}$ and $D_{R+1} = D_R \cup \{R + 1\}$. The rounding bound implies $s_R/4^R \rightarrow \delta > 0$. Finally, use $q_{M+2}(d) = 4q_M(d) + 21\mathbf{1}_{d \mid M+1} + \mathbf{1}_{d \mid M+2}$, together with the analogous target identity and $q_{2R+2}(R + 1) = 2^{R+1} + 1$, to obtain (3). The full comparison induction and both quotient identities are given in [the companion paper's integer-quotient proof for \$1/21\$](#) . $\square$

The finite uniqueness statement is also explicit: if $D \subseteq \{2, \dots, R\}$ and an integer s satisfy $\sum_{d \in D} q_{2R}(d) + s = T_{2R}$ with $0 \leq s \leq 2^R$, then $D = D_R$ and $s = s_R$. This is the denominator-specific separation theorem ([uniqueness of a finite representation with the stated remainder bound](#)). The cited finite crossing lemmas give additional consequences under their alignment hypotheses: an earlier finite prefix cannot occur, and a real greedy exponent must be skipped ([the missing-prefix consequence of an aligned crossing](#), [the real greedy skip forced by that crossing](#)).

Approximation to $1/2$ by finite supports. For $A \subseteq \{2, \dots, M\}$ and $1 \leq m \leq M$, define the integer

$$K_A(m) = 2^{m-1} - \sum_{j=2}^m 2^{m-j} c_A(j).$$

It measures the error in the truncated divisor-coefficient sum, after multiplication by 2^m . In particular, $K_A(1) = 1$ and $K_A(m+1) = 2K_A(m) - c_A(m+1)$.

Theorem D.2 (finite approximations with vanishing scaled error). *Suppose there are integers $M_j \geq 1$ tending to infinity and sets $A_j \subseteq \{2, \dots, M_j\}$ such that*

$$\frac{|K_{A_j}(M_j)|}{2^{M_j}} \rightarrow 0.$$

Then $X_A(2) = 1/2$ for some infinite set $A \subseteq \mathbb{N}_{>0}$.

To compare the finite subseries with $1/2$, we separate the coefficient tail from the terminal carry. No agreement between the supports A_j or bound on their earlier carries is needed. The estimate is

$$\left| X_{A_j}(2) - \frac{1}{2} \right| \leq \frac{|K_{A_j}(M_j)| + 2\sqrt{M_j} + 4}{2^{M_j}}.$$

Indeed, $c_{A_j}(n) \leq \tau(n) \leq 2\sqrt{n}$ and $\sqrt{M+r} \leq \sqrt{M} + r$, so the normalised tail is at most $\sum_{r \geq 1} 2^{-r} (2\sqrt{M} + 2r) = 2\sqrt{M} + 4$. The finite sums therefore tend to $1/2$. The achievement set is closed, and no finite support has value $1/2$, so its representing support is infinite. A square-root bound on the terminal carry would suffice, but the theorem permits any error $o(2^{M_j})$. For a finite $D \subseteq \{2, \dots, M\}$, the floor-quotient identity gives

$$K_D(M) = 2^{M-1} - \sum_{d \in D} \left\lfloor \frac{2^M}{2^d - 1} \right\rfloor.$$

The exact quotient rows considered in the companion paper satisfy $\sum_{d \in D} q_M(d) = 2^{M-1} - 1$, so they have $K_D(M) = 1$. The terminal criterion allows a larger error of either sign. At a fixed depth these are different conditions, although cofinal existence of either kind of approximation characterises $1/2 \in A$.

Conversely, if $X_A(2) = 1/2$, then $1 \notin A$, and the prefixes $A_M = A \cap \{2, \dots, M\}$ satisfy

$$K_{A_M}(M) = \sum_{r \geq 1} c_A(M+r) 2^{-r} \leq 2\sqrt{M} + 4.$$

The equality uses the coefficients of the full support A on the right: truncation does not change the coefficients through M . Hence the existence hypothesis of Theorem D.2 is equivalent to $1/2 \in A$. No compatible sequence of prefixes is required for the forward implication. The existence of the approximating supports remains unproved and would refute Problem 1.1.

Finite families with a shared prefix. Put $B(m) = 2\lfloor \sqrt{m} \rfloor + 4$. For $0 \leq K \leq M$, consider a family $A_1, \dots, A_{B(M)} \subseteq \{2, \dots, M\}$ with

$$1 \leq K_{A_k}(m) \leq B(m) \quad (1 \leq m \leq M), \quad K_{A_k}(M) = k.$$

Require that all the A_k agree on $\{1, \dots, K\}$ and that some integer $E \geq B(M)$ satisfies

$$\sum_{j=K+1}^M 1_{A_k}(j) 2^{M-j} + k = E \quad (1 \leq k \leq B(M)).$$

Thus the binary suffixes have consecutive values $E - k$, while the terminal carry ranges through the entire permitted interval. This is substantially more data than one finite approximation to $1/2$.

Theorem D.3 (unbounded shared-prefix families represent one half). *Suppose that for every N there are M, K with $\max\{N, 1\} \leq M$, $0 \leq K \leq M$, and a family satisfying all the conditions in the preceding paragraph. Then $X_A(2) = 1/2$ for some infinite set $A \subseteq \mathbb{N}_{>0}$.*

For each M choose the member with terminal carry 1. Its error 2^{-M} tends to zero along unbounded depths, so Theorem D.2 applies. The other shared-prefix conditions are additional requirements of the proposed construction; the compactness argument uses only this one member. No construction at unbounded depths is established here.

The tests above leave three kinds of arithmetic information to be supplied: returns of the actual greedy sequence, finite approximants at unbounded depths, or an exclusion of a possible final skip. The following questions specify those missing inputs.

Problem D.4 (membership of $1/21$ in the Mersenne achievement set). For the greedy remainder of $1/21$, prove that arbitrarily large N satisfy

$$2^{N+1} r_N(1/21) < \frac{2^{N+1}}{2^{N+1} - 1}, \quad \text{equivalently} \quad r_N(1/21) < \frac{1}{2^{N+1} - 1}.$$

Equivalently, exclude the condition $\mathcal{F}_{21}$ defined above. One sufficient route is to rule out the eventual recurrence in Theorem D.1(3) together with $s_R > 2^R$; another is to prove $s_R \leq 2^R$ at arbitrarily large ranks. Neither sufficient route is claimed to be necessary by itself.

Problem D.5 (bounded returns of the scaled remainder). For each of the targets $x = 1/2$ and $x = 1/21$, does the greedy remainder $r_N(x)$ return to one bounded interval after scaling by 2^N ?

$$\exists B < \infty \forall K \exists N \geq K : \quad 2^N r_N(x) \leq B.$$

Problem D.6 (arithmetic tests for the greedy sequence). Can a finite-memory, 2-adic or discrepancy argument prove $s_R \leq 2^R$ at arbitrarily large ranks, or rule out the eventual recurrence in Theorem D.1(3)? Such an argument must use the divisor counts of the actual greedy support. Can a bounded window of $R \bmod 6$, residues of s_R , endpoint divisor counts and the finite set of eventual skips force a decrease or a contradiction? Alternatively, can one show that these bounded-memory data cannot distinguish the actual sequence from sequences that remain above 2^R but need not come from a support?

Problem D.7 (final-skip Diophantine exclusion). Let $E = \sum_{n \geq 1} (2^n - 1)^{-1}$. If $1/21 \notin A$, let M be the last skipped exponent, S_M its finite skipped prefix, and

$$a_M = \frac{1}{21} + \sum_{d \in S_M} \frac{1}{2^d - 1}.$$

Write $\text{gap}_M = (2^M - 1)^{-1} - R_M > 0$. Can the arithmetic restrictions on this last skipped exponent be used to prove $|E - a_M| \geq \text{gap}_M$, contradicting

$$0 < a_M - E < \text{gap}_M?$$

E Elementary details used in the proofs

Cyclotomic prime-power fact. Let $b \geq 2$, $n \geq 2$, $\ell \mid \Phi_n(b)$ be prime and $e = v_\ell(b^n - 1)$. Then $\text{ord}_{\ell^e}(b) = n$. We include the 2-adic case. For odd ℓ , put $d = \text{ord}_\ell(b)$ and $s = v_\ell(b^d - 1)$. The elementary lifting identity $v_\ell(b^{dt} - 1) = s + v_\ell(t)$ follows by factoring a geometric sum when $\ell \nmid t$ and by a binomial expansion for a factor ℓ . In the factorisation $b^n - 1 = \prod_{r \mid n} \Phi_r(b)$, subtracting these valuations over proper divisors gives

$$v_\ell(\Phi_n(b)) = \begin{cases} s & n = d, \\ 1 & n = d\ell^j, j \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

If $n = d$, no smaller positive exponent gives $b^m \equiv 1 \pmod{\ell}$. If $n = d\ell^j$, the full valuation is $e = s + j$; the same lifting identity shows that the least exponent giving valuation e is $d\ell^j = n$. For $\ell = 2$, b is odd. Factoring $b^{2^j} - 1$ gives

$$v_2(b^{2^j} - 1) = v_2(b - 1) + v_2(b + 1) + j - 1 \quad (j \geq 1).$$

The only possible indices $n \geq 2$ are $n = 2^j$: $v_2(\Phi_2(b)) = v_2(b + 1)$ and $v_2(\Phi_{2^j}(b)) = 1$ for $j \geq 2$; all other indices have valuation zero. The displayed identity again makes 2^j the least exponent with the full valuation. This proves the fact. Finally $\Phi_n(b) > 1$: each primitive n th root ζ satisfies $|b - \zeta| > 1$, and their product is $\Phi_n(b)$.

Exact scalar cover cost. For $t \geq 1$, put $z = \log_2 t$. The substitution $y = 2^z$ reduces the infimum defining $\Psi(t)$ to $y^z/(y - 1)$ on $1 < y \leq 2$. Its derivative has the sign of $(z - 1)y - z$. Hence

$$\Psi(t) = \begin{cases} t, & 1 \leq t \leq 4, \\ z^z/(z - 1)^{z-1}, & t > 4. \end{cases}$$

The endpoint and the open lower bound $\alpha > 0$ are both included in this calculation. This scalar identity does not supply a converse to the positive-cover criterion.

The cost for a finite set of divisors

For the following finite divisor sets, the lower bound is asymptotically attained. Here $K_*(F)$ is the infimum of the cover cost K from Section 5, taken over finite or countable covers of F and all positive weights of total one. For $F(q, P) = \{qd : d \mid \prod_{p \in P} p\}$, where $q \geq 2$ and no $p \in P$ divides q , put $S = \sum_{p \in P} 1/p$. If $S \geq 1$, then

$$\frac{e(S-1)}{q} \leq K_*(F(q, P)) \leq \frac{eS}{q}. \quad (13)$$

For the lower bound, we condition on $q \mid n$ and write $f_F(n) = 2^Z$. The Chinese remainder theorem gives $\mathbb{E}Z = S$. For $z \geq 0$ and $v = \alpha \log 2 > 0$,

$$\frac{2^{\alpha z}}{2^\alpha - 1} \geq \frac{e^{(z-1)v}}{v} \geq e(z-1) \quad (z > 1);$$

for $0 \leq z \leq 1$ the claimed lower bound is nonpositive. Thus $\Psi(2^z) \geq e(z-1)$. For the upper bound, use the one-set cover, whose weight is one, with $z = 1/S$ and $\alpha = \log_2(1+z)$; its exact positive expansion has cost

$$\frac{1}{qz} \prod_{p \in P} (1 + z/p) \leq \frac{eS}{q}.$$

In particular, $1 - 1/S \leq qK_*(F(q, P))/(eS) \leq 1$. Thus $K_*(F(q, P)) \sim eS/q$ as $S \rightarrow \infty$, uniformly over the permitted choices of q and P .

F Sources, verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

For Erdős's reciprocal-summable extension and fractional-part argument, see pp. 222 and 226 of [7]. The periodic theorem [12] is also stated in Luca and Tachiya's account [11, Theorem A and Example 2, pp. 139–140], and Hornich's strict-tail theorem is proved in Nitecki's exposition [10, Theorem 4(1)]. These expositions are the sources used here for those two results; the original articles were not independently retrieved. The *Formal Conjectures* file [6] is statement-level prior art, not a proof dependency.

The evidence record identifies the exact statements, source revisions and recorded comparisons. The mixed implication has a formal proof. A separate formal construction of a separating support V uses finite sets of squarefree divisors. The A_* calculation, printed fresh-prime construction, Corollary 5.2, signed finite-denominator extension

and Theorem D.1 have ordinary arguments here, without independent human review. The strictness corollary has no formal binding in the record. The companion paper locates the separate comparisons for its three logarithmic-sampling results. No historical priority is asserted for the support comparison.

Reproducing the weighted theorem. The two declarations for Theorem 1.2 are in the public source at [commit 91ca3405](#). That snapshot pins Lean 4.29.1 in `lean-toolchain` and its Mathlib revision in `lake-manifest.json`. From a complete checkout, with `elan` installed, build the two modules with:

```
lake exe cache get
python3 scripts/lean_fast_build.py --jobs 2 \
  ErdosProblems.Erdos257.PaperCompleteR8.WeightedReturn \
  ErdosProblems.Erdos257.PaperCompleteR8.WeightedHereditaryClaim
```

The cache download is optional. The modules contain `divisibilityWeightedClaim` and `finitePrimeWeighted_fixedBase_hereditary`, respectively; their precise statements and associated checks are linked above.

The supplementary declarations are collected in [the companion paper’s final source section](#), grouped by support criteria, finite denominators, integer recurrences and achievement sets.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.2	Lean · Comparator	Theorem 3.1	Lean · Comparator
Theorem 4.1	Lean · Comparator	Proposition 5.1	Lean · Comparator pending
Corollary 5.2	Verification record	Theorem A.1	Lean · Comparator
Theorem B.1	Lean · Comparator	Theorem C.1	Lean · Comparator
Theorem D.1	Lean · Comparator	Theorem D.2	Lean · Comparator pending
Theorem D.3	Lean · Comparator		